

THE UNITED REPUBLIC OF TANZANIA
NATIONAL EXAMINATIONS COUNCIL OF TANZANIA
CERTIFICATE OF SECONDARY EDUCATION EXAMINATION

036/1

INFORMATION AND COMPUTER STUDIES 1
(For Both School and Private Candidates)

Year: 2025

Duration: 3 Hours

Instructions

1. This paper consists of sections A, B, and C with a total of **eleven (11)** questions.
2. Answer **all** questions in sections A and B, and **two (2)** questions from section C.
3. Section A carries **sixteen (16)** marks, section B **fifty four (54)** marks and section C carries **thirty (30)** marks.
4. Communication devices and any other unauthorized materials are **not** allowed in the examination room.
5. Write your **Examination Number** on every page of your answer booklet(s).



(vii)

SECTION A (16 Marks)
Answer **all** questions in this section.

1. For each of the items (i) - (x), choose the correct answer from the given alternatives and write its letter beside the item number in the answer booklet provided.

(i) Which menu option will you use to display the ruler in Microsoft word?

- A View B Insert C Home
D Review E ,Layout
- (viii)

(ii) Which cybercrime did the criminal commit when tapping communication channel to get information secretly?

- A Fraud B Identity theft C Denial of Service
D Phishing E Eaves dropping
- (ix)

(iii) What type of the computer have a large storage capacity and allows hundreds of customers to work on the same data at the same time?

- A Super computer B Minicomputer C Microcomputer
D Mainframe computer E Personal computer

(iv) How can you differentiate flat file database from relational database?

- A Flat file database holds many records while relational database holds one record.
B Flat file database offers strong reporting compared to relational database.
C Flat file database import data from other software while relational database did not.
D Flat file database contains single table but relational database includes many tables.
E Flat file database contains many tables while relational database includes a single table.

(v) What will be the output after executing the following HTML codes?

```
<html>
  <head><title> </title></head>
    <body>
      <a href="url"> Tanzania </a>
    </body>
</html>
```

- A Tanzania B Tanzania
C Tanzania D *Tanzania*
E "Tanzania"

(vi) Which type of text alignment would you use to align texts evenly along the left and right margins in a word document?

- A Distributed alignment B Left alignment C Center alignment
D Right alignment E Justified alignment

(vii) Which one are the correct procedures for adding a chart in one of the slides in PowerPoint?

- A Insert>Chart>Picture D Picture>Insert>Chart
B Picture>Chart>Insert E Chart>Insert>Picture
C Insert>Picture>Chart

(viii) Which type of a chart will be most appropriate for comparing performance of sales of two items?

- A Pie chart B Column chart C Line chart
D Bubble chart E Surface chart

(ix) How would you protect the computer from malware attacks?

- A By purchasing the original antivirus software
B By scanning removable devices before opening them
C By opening email attachments from unknown sources
D By visiting any websites in the World Wide Web
E By switching off the computer regularly

(x) Which type of multimedia is used in a computer and smartphone games?

- A Non-linear multimedia B Digital multimedia
C Linear multimedia D Visual multimedia
E Analog multimedia

2. Match the functions of field properties in **List A** to their corresponding names of the field properties from Microsoft Access in **List B** by writing the letter of the correct response beside the item number in the answer booklet provided.

List A	List B
(i) It allows the user to set the type of numbers to be entered in a field. ✕	A Validation Rule
(ii) It determines if an entry must be made in the field before you proceed to the next field or record. ^/	B Indexed
(iii) It restricts the values to be entered in a field.	C Default Value
(iv) It formats automatically the field entry into a specific pattern.	D Caption
(v) It allows a message to be displayed once the validation rule is violated. ✕	E Validation Text
(vi) It describes the label for the field to be used in a table or a form view.	F Field size
	G Input Mask
	H Required
	I Format

SECTION B (54 Marks)
Answer **all** questions in this section.

3. (a) Why is it advised to close windows and doors in a computer laboratory? Give four reasons. (6 marks)
- (b) What are the two effects of opening windows and doors on computers in a laboratory? (3 marks)
4. (a) Why is it prohibited to use liquid fire extinguisher in the computer laboratory? (1 mark)
- (b) You have been asked to prepare a presentation on the health's problems that a person may suffer when using ICT devices for a long time. (4 marks)
- (i) Explain two healthy problems that you would include in your presentation. (4 marks)
- (ii) Explain two protective ways for each problem explained in part (b) (i). (4 marks)
5. (a) Why network administrators in organizations prefer to install Linux Operating System in their server computers? (3 marks)
- (b) What two possible challenges the organization may face when using the Linux operating system? (2 marks)
- (c) Explain two functions of the Linux Operating System used in organization's Server Computers. (4 marks)
6. Form Six students of a certain school plans to capture a video in their graduation ceremony. However, they lacked knowledge of processing their video and they asked you to assist them in editing the videos before compiling in one Flash disk. (5 marks)
- (a) Explain five procedures you would use to accomplish the task. (5 marks)
- (b) Explain two types of compression method you may use to accomplish the goal. (4 marks)
7. You have been assigned by the ICS teacher to prepare the students' magazine and submit it to his/her email addresses. When composing it, three fields named *To* field, *CC* field, and *BCC* field appeared. (6 marks)
- (a) Why the teacher opted such way of communication? Briefly explain by giving three reasons. (6 marks)
- (b) Explain the function of each field appeared during composing an e-mail. (3 marks)
8. The Ministry of Health uses Television and Radio broadcasting to disseminate information about pandemic diseases and measures to be taken by the society. (1 mark)
- (a) Which type of information dissemination method used by the Ministry of health? (1 mark)
- (b) Why the Ministry uses such means of disseminating information? Give four reasons. (6 marks)
- (c) What two challenges would the Ministry face by using such method of disseminating information? (2 marks)

SECTION C (30 Marks)

Answer **two (2)** questions from this section.

9. Why most of newspapers developers prefer to use desktop publishers than other programs?
Explain by giving five points. **(15 marks)**
10. Recently, most organizations opt to use the computer technology instead of human being in their various operations. Using five points, explain the reasons for their choices. **(15 marks)**
11. Explain five advantages of using computers and other devices which are connected to the network in an organization. **(15 marks)**

3-25

7. the flash disk.

6. b) Lossy compression is
Lossless -

- reduces file size by removing redundant data.
- compresses files without losing quality.

7. b) Main receiver of the email.

- sends the copy to other recipients who can see each other.

**THE UNITED REPUBLIC OF TANZANIA
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036/2

**INFORMATION AND COMPUTER STUDIES 2
(ACTUAL PRACTICAL)
(For Both School and Private Candidates)**

Duration: 3 Hours

Year: 2025

Instructions

1. This paper consists of **three (3)** questions.
2. Answer any **two (2)** questions. Each question carries **25** marks.
3. Save your work on the desktop using the folder named by your examination number.
4. Save your work by using the 1997-2003 version of the MS Office software you are using.
5. Check whether the **printed** work(s) are similar to the **softcopy** saved in the folder.
6. Submit printed hardcopy together with the softcopy of your work.
7. Cellular phones and any unauthorised materials are **not** allowed in the examination room.
8. Write your **Examination Number** on every page of your softcopy work.



1. Mr. Majao manages his Shoes Shop records using Microsoft Access database. The database stores customer records including contacts and purchases. He uses the database as the marketing strategy for facilitating him in awarding his royal customers. The customer awarded a free pair of shoes if he buys 10 pairs. Study carefully the following tables which are included in the database and answer the questions that follow:

Customers

CustID	FirstName	Surname	Phone#
1	Jona	Muliye	0713345610
2	Mary	Kolida	0674234012
3	Kilo	Kilosa	0782919293
4	Juju	Moses	0761839099
5	Marcelina	Mosi	0651828280
6	Moneylisa	Pesa	0621345334
7	Joy	Majiji	0765080911

Purchases

PurchaseID	CustID	DatePurchase	Shoe_Category	Num_Shoes	Amount
1	1	3/8/2023	ladies	1	35,000.00
2	1	4/6/2023	ladies	3	30,000.00
3	1	5/5/2023	kids	4	60,000.00
4	1	4/7/2023	Men	2	48,000.00
5	2	8/2/2023	ladies	2	30,000.00
6	2	5/23/2023	kids	1	15,000.00
7	3	6/23/2023	men	1	39,000.00
8	3	6/18/2023	kids	2	28,000.00
9	4	3/26/2023	men	2	56,000.00
10	5	4/6/2023	kids	3	33,000.00
11	5	5/20/2023	ladies	2	40,000.00
12	6	4/23/2016	men	1	35,000.00
13	7	5/23/2023	ladies	1	18,000.00
14	7	5/27/2023	ladies, men	4	98,000.00
15	7	6/20/2023	kids, ladies, men	3	85,000.00
16	7	6/28/2023	kids	2	30,000.00

- Create the database for Mr. Majao Shoes Shop and save it as "MajaoShoesDB".
- Create the given Customers and Purchases tables within MajaoShoesDB.
- Use Form Wizard to create a form that enables Mr. Majao to enter customers and purchases data using one form. Use the form created to enter the sample data provided. Save the form as "CustomersPurchasesform".
- Use Customers and Purchases tables to create a one-to-many relationship between the two tables.
- Add mathematical formula to the form created in 1 (c) that enables Majao to view the total amount of money that each customer spent and the number of pairs bought.

- (f) Create a query that enables to check customers' purchase details. The details include customer ID, first name, surname, Phone number, date purchased, pairs of shoes and amount paid for each day. Save the query as "customerspurchases".
- (g) Create a query which enables to view customer's first name, surname, dates and shoes category for customers who bought ladies or kids shoes. Save the query as "bought ladies or kids shoes".
- (h) Create a parameter query that enables to search for any customer using customer's surname and display his purchase records. Save the query as "search a customer by surname".
- (i) Create a report that search and prints customers' purchase records by their surnames. Save the report as "Report of Customers by surname".
- (j) Create a report that will enable Majao to print records of all customers who bought men shoes. Save the report as "Men shoes only report".

(25 marks)

2. Umoja Technical College offers different technical courses in its campuses allocated at different areas in Tanzania. To make teachers and students aware of all monthly-scheduled events, the principal of the college asked you to prepare online calendar to the college website.

- (a) Create the following Interface by using basic HTML codes.

Hint: Use table with border 0, height 300 and width 300 to create the Interface and to arrange the monthly buttons.

UMOJA CALENDAR		
January	February	March
April	May	June
July	August	September
October	November	December

Page descriptions

- The heading “UMOJA CALENDAR” should be colored in red with the font type “Gabriola” and text size 20.
- The size of the horizontal line should be 300.

(b) Activate the button “May” in the interface created in 2 (a) so that when the user clicks, it displays the following page.

May 2025						
Sun	Mon	Tue	Wed	Thur	Fri	Sat
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	31			

HINT:

- Use Microsoft word to create the circle and rectangle shapes to the dates “1” and “4” respectively.
- Use the borderless table to arrange the dates.

Page descriptions

- The heading “May 2025” should be in level 2.
- The size of the horizontal line after the heading should be 300 and aligned at centre.
- The background and line colour of the circle shape should be orange, accent = 2, lighter = 80% and the line colour should be blue.
- The line colour of the rectangle shape in the 4th date should be blue.
- The height and width of the table required to arrange the dates should be 350 for each.

(c) Develop the HTML codes that activate the 1st date in the circle shape created in 2(b) such that the following page will be displayed when the user clicks it.

1st May 2025

Worker's day

Page descriptions

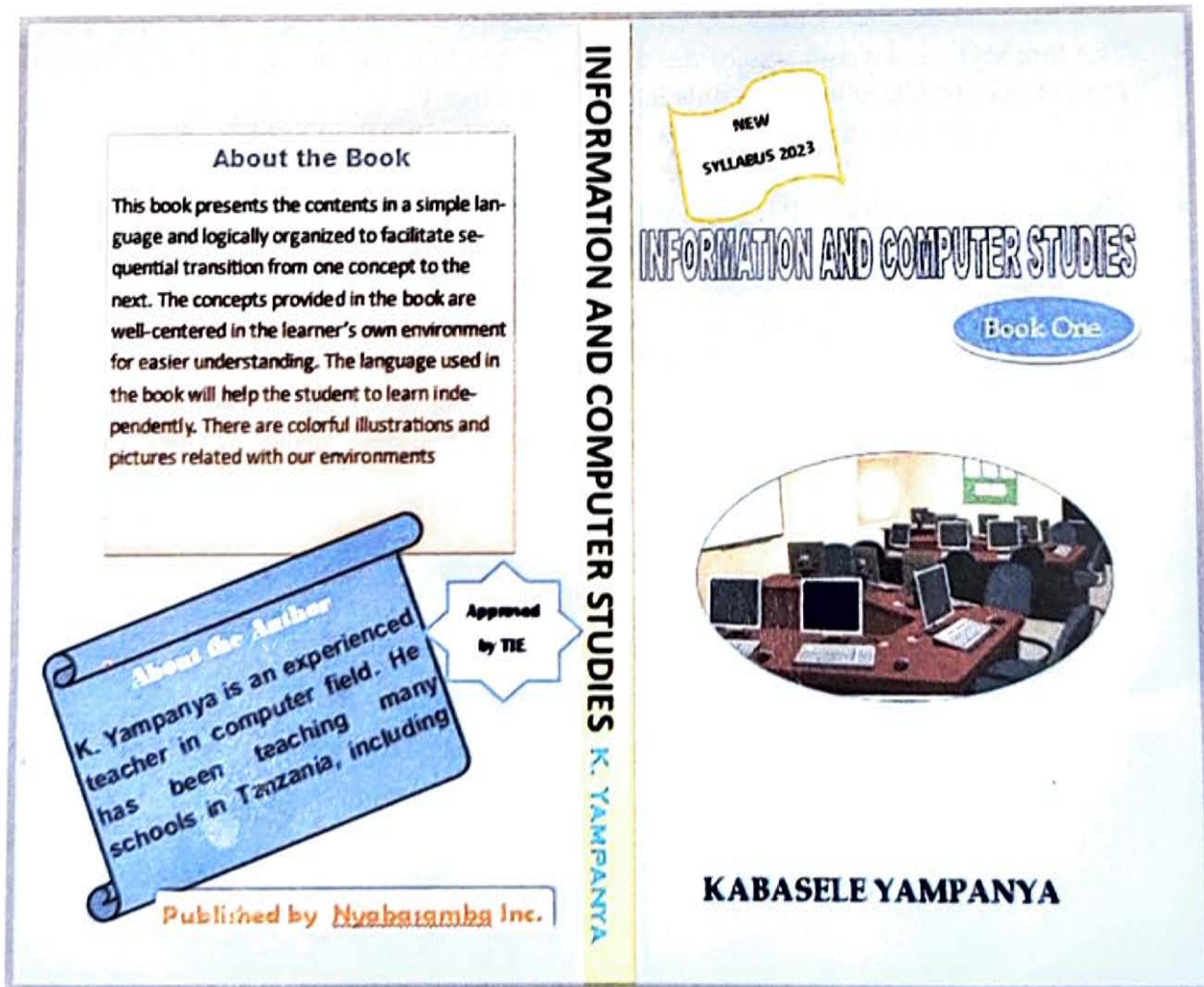
- The font type, colour and size of the text “Worker’s day” should be “Lucida handwriting”, “white” and 13 respectively and the background color should be green.

- (d) Construct HTML codes that display the “current date” when the user hangs the mouse on the rectangle button labeled with number “4” in the interface created in 2 (b).

(25 marks)

3. Mr. Kabasele Yampanya prepared the Form One ICS book using a new syllabus. However, he lacks knowledge of preparing the book cover page. You are required to design the following cover page.

HINT: Use the Microsoft Publisher in accomplishing the task.



Cover Page descriptions

- The background color of the whole page should be 30% tint of accent 1.
- The font style, color and size of the title "About the book" should be Castellar, light blue and 16 respectively.
- The font style and size of the book descriptions below the title "About the book" should be Calibri and 14 respectively with the background color: Accent 2, lighter 40%.
- The font style, color and size of the title "About the Author" should be Bodoni MT Black, white and 16 respectively.
- The font style and size of the author descriptions below the title "About the Author" should be Arial and 16 respectively.
- The background color of the vertical scroll shape on "About the Author" should be blue with black lines borders.
- The font style and size of the text "Approved by TIE" should be Britannic Bold and 12 respectively.
- The background color of the star shape on "Approved by TIE" should be Outline-white, Accent 1 with blue lines borders.
- The font style, color and size of the text "Published by Nyabasamba Inc." should be Berlin Sans FB Demi, Accent 2 (RGB) and 14 respectively.
- The font style and size of the text "INFORMATION AND COMPUTER STUDIES" at the middle of the cover page should be Bahnschrift SemiCondensed, 26 respectively with background color Linear Up Gradient – Accent 4.
- The font style, color and size of the text "K. YAMPANYA" at the middle of the cover page should be Castellar, light blue and 14 respectively.
- The font style and size of the text "New Syllabus 2023" should be Calibri and 12 respectively.
- The background color of the shape on the text "New Syllabus 2023" should be Outline-white, Accent 1 with gold lines borders.
- The font style and size of the text "Book One" should be Book Antiqua and 18 respectively with background color fill Accent 1.
- The style of the word "INFORMATION AND COMPUTER STUDIES" should be Word Art: Fill-Cream, Outline-blue.
- The font style and size of the texts "KABASELE YAMPANYA" should be Castellar and 20 size.

(25 marks)

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036/1

INFORMATION AND COMPUTER STUDIES 1

(For Both School and Private Candidates)

Duration: 3 Hours

SOLUTIONS

Year: 2025

Instructions

1. This paper consists of sections A, B and C with a total of **eleven (11)** questions.
2. Answer **all** questions in section A and B and **two (2)** questions from section C.
3. Section A carries **sixteen (16)** marks, section B **fifty four (54)** marks and section C **thirty (30)** marks.
4. Communication devices and any unauthorised materials are **not** allowed in the examination room.
5. Write your **Examination Number** on every page of your answer booklet(s)

SECTION A (16 Marks)

Answer all questions in this section.

1. For each of the items (i) to (x), choose the correct answer from the given alternatives and write its letter beside the item number.

(i) Which menu option will you use to display the ruler in Microsoft Word?

- A. Home
- B. Insert
- C. View
- D. Review

Correct answer: C. View

The ruler in Microsoft Word is found under the View menu. This menu controls elements related to how the document appears on the screen, including rulers, gridlines, and zoom options.

(ii) Which cybercrime did the criminal commit when tapping a communication channel to get information secretly?

- A. Fraud
- B. Identity theft
- C. Eavesdropping
- D. Phishing

Correct answer: C. Eavesdropping

Eavesdropping refers to secretly listening to or intercepting communication without authorization. Tapping communication channels to obtain information fits this definition.

(iii) What type of computer has a large storage capacity and allows hundreds of customers to work on the same data at the same time?

- A. Supercomputer
- B. Mainframe computer
- C. Personal computer
- D. Minicomputer

Correct answer: B. Mainframe computer

Mainframe computers are designed to support many users simultaneously. They have large storage capacity and are commonly used in banks, universities, and large organizations.

(iv) How can you differentiate a flat file database from a relational database?

- A. Flat file database imports data from other software while relational database does not
- B. Flat file database contains a single table but relational database includes many tables
- C. Flat file database contains many tables while relational database includes a single table
- D. Flat file database offers stronger reporting than relational database

Correct answer: B

A flat file database stores all data in one table, which limits flexibility. A relational database uses multiple related tables, making it more efficient and organized for large datasets.

(v) What will be the output after executing the following HTML codes?

- A. Tanzania
- B. Tanzania

C. Home

D. url

Correct answer: B. Tanzania

The anchor tag displays the text between the opening and closing tags on the webpage. The word “Tanzania” appears as a clickable link.

(vi) Which type of text alignment would you use to align texts evenly along the left and right margins in a Word document?

A. Left alignment

B. Center alignment

C. Right alignment

D. Justified alignment

Correct answer: D. Justified alignment

Justified alignment ensures that text is evenly aligned on both the left and right margins, giving documents a clean and professional appearance.

(vii) Which one are the correct procedures for adding a chart in one of the slides in PowerPoint?

A. Insert > Chart > Picture

B. Picture > Chart > Insert

C. Insert > Picture > Chart

D. Picture > Insert > Chart

Correct answer: C. Insert > Picture > Chart

Charts in PowerPoint are added through the Insert menu. This allows users to select and insert different types of charts into slides.

(viii) Which type of chart will be most appropriate for comparing performance of sales of two items?

- A. Pie chart
- B. Column chart
- C. Bubble chart
- D. Surface chart

Correct answer: B. Column chart

Column charts are effective for comparing quantities between different items because the height of the columns makes differences easy to see.

(ix) How would you protect the computer from malware attacks?

- A. By opening email attachments from unknown sources
- B. By visiting any websites in the World Wide Web
- C. By switching off the computer regularly
- D. By scanning removable devices before opening them

Correct answer: D

Scanning removable devices such as flash disks helps detect malware before it spreads into the computer system.

(x) Which type of multimedia is used in computer and smartphone games?

- A. Linear multimedia
- B. Non linear multimedia
- C. Analog multimedia
- D. Visual multimedia

Correct answer: B. Non linear multimedia

Games allow users to interact and make choices that affect outcomes. This interaction makes them non linear multimedia.

2. Match the functions of field properties in List A with their corresponding names of the field properties from Microsoft Access in List B.

List A

- (i) It allows the user to set the type of numbers to be entered in a field.
- (ii) It determines if an entry must be made in the field before proceeding to the next field or record.
- (iii) It restricts the values to be entered in a field.
- (iv) It formats automatically the field entry into a specific pattern.
- (v) It allows a message to be displayed once the validation rule is violated.
- (vi) It describes the label for the field to be used in a table or form view.

List B

- A. Validation Rule
- B. Indexed
- C. Default Value
- D. Caption
- E. Validation Text
- F. Field Size
- G. Input Mask
- H. Required

Answers.

- (i) F (ii) H (iii) A (iv) G (v) E (vi) D

SECTION B (54 Marks)

Answer all questions in this section.

3. (a) Why is it advised to close windows and doors in a computer laboratory? Give four reasons.

One reason is to prevent dust from entering the laboratory. Dust can accumulate inside computer components and cause overheating or damage.

Another reason is to protect computers from moisture. Rainwater and humidity can cause short circuits and damage electronic parts.

Closing windows and doors also improves security. It reduces the risk of theft or unauthorized access to computer equipment.

It also helps maintain a controlled temperature. Computers perform better in stable temperature conditions.

- (b) What are the two effects of opening windows and doors on computers in a laboratory?

Opening windows allows dust to enter the laboratory, which can damage internal components of computers.

It may also allow moisture to enter, leading to rusting, electrical faults, or system failure.

4. (a) Why is it prohibited to use liquid fire extinguishers in the computer laboratory?

Liquid fire extinguishers conduct electricity and can damage computer equipment. They also increase the risk of electric shock to users.

(b) Explain two health problems a person may suffer when using ICT devices for a long time.

One health problem is eye strain. Continuous exposure to screens causes tired eyes, headaches, and blurred vision due to excessive screen brightness.

Another problem is back pain. Sitting for long hours with poor posture puts pressure on the spine and muscles.

(c) Explain two protective ways for each problem explained in part (b).

To prevent eye strain, users should take regular breaks and adjust screen brightness to a comfortable level.

To reduce back pain, users should use ergonomic chairs and maintain proper sitting posture.

5. (a) Why network administrators in organizations prefer to install Linux operating system in their server computers?

One reason is stability. Linux servers rarely crash, making them reliable for continuous operation.

Another reason is security. Linux is less vulnerable to viruses and malware compared to other operating systems.

Administrators also prefer Linux because it supports multiple users efficiently.

(b) What two possible challenges the organization may face when using Linux operating system?

One challenge is limited availability of some commercial software designed mainly for other operating systems.

Another challenge is the need for skilled personnel to manage and maintain Linux systems.

(c) Explain two functions of the Linux operating system used in organization servers.

Linux manages hardware resources efficiently, ensuring optimal use of memory and processing power.

It also controls user access and security, protecting organizational data.

6. Form Six students plan to capture a video in their graduation ceremony but lack knowledge of video processing.

(a) Explain five procedures you would use to accomplish the task.

The first procedure is importing video files into video editing software.

The second procedure is trimming unwanted sections to remove errors.

The third procedure is arranging video clips in the correct sequence.

The fourth procedure is adding transitions, text, and effects to enhance quality.

The final procedure is exporting the edited video to a flash disk.

(b) Explain two types of compression method you may use.

Lossy compression reduces file size by removing some data, slightly affecting quality.

Lossless compression reduces file size without losing any data.

7. You have been assigned to prepare the students' magazine and submit it through email.

(a) Why did the teacher opt for this way of communication? Give three reasons.

Email is fast and allows instant delivery of information.

It is cost effective because it does not require printing or physical transport.

It allows sending the magazine to many recipients at the same time.

(b) Explain the function of each field appeared during composing an email.

The To field contains the main recipient of the email.

The CC field sends a copy to additional recipients who are allowed to see others.

The BCC field sends copies without revealing recipients' addresses.

8. The Ministry of Health uses television and radio broadcasting to disseminate information about pandemic diseases.

(a) Which type of information dissemination method is used by the Ministry of Health?

The method used is mass media communication.

(b) Why does the Ministry use such means of disseminating information? Give four reasons.

Mass media reaches a large population at the same time.

Information is delivered quickly during emergencies.

It reaches both literate and illiterate people.

It covers wide geographical areas.

(c) What two challenges would the Ministry face by using such method?

One challenge is high operational costs.

Another challenge is limited feedback from the audience.

SECTION C

Answer two (2) questions from this section.

9. Why do most newspaper developers prefer to use desktop publishing programs than other programs? Explain by giving five points.

Their ability to produce professional layouts. Desktop publishing software such as Adobe InDesign and Microsoft Publisher allows precise control over text, images, columns, margins, and spacing, which is essential for producing attractive and readable newspapers.

Flexibility in text and image editing. Desktop publishing programs allow developers to easily edit text, resize images, adjust fonts, and apply styles without affecting the overall layout, making corrections and updates faster and more efficient.

Desktop publishing programs also support high quality printing. They are designed to work with printing standards such as color separation and resolution control, ensuring that newspapers are printed clearly and professionally.

Time efficiency. These programs provide ready made templates and tools that speed up the design process, allowing newspapers to be produced within tight deadlines.

Lastly, desktop publishing programs support integration of graphics. Developers can easily import photographs, charts, and illustrations, enhancing the visual appeal and communication effectiveness of newspapers.

10. Recently, most organizations opt to use computer technology instead of human beings in their various operations. Using five points, explain the reasons for their choices.

One major reason organizations use computer technology is

Speed. Computers can process large amounts of data much faster than humans, enabling organizations to complete tasks quickly and meet operational deadlines.

Accuracy. Computers reduce errors in calculations, data processing, and record keeping, leading to more reliable results compared to manual work.

Cost reduction is also a reason for adopting computer technology. Although initial investment may be high, computers reduce long term costs by minimizing labor expenses and increasing productivity.

Computers also allow efficient data storage and retrieval. Large volumes of information can be stored digitally and accessed easily whenever needed, improving decision making and organizational efficiency.

Finally, computer technology enables automation. Routine and repetitive tasks can be automated, allowing human workers to focus on more complex and creative activities.

11. Explain five advantages of using computers and other devices which are connected to the network in an organization.

One advantage of using networked computers is

Improved communication. Employees can share information quickly through emails, messaging systems, and shared platforms, enhancing coordination within the organization.

Resource sharing. Devices connected to a network can share printers, files, and internet connections, reducing costs and improving efficiency.

Networked computers also improve data management. Centralized databases allow secure storage, easy updating, and controlled access to organizational information.

Enhanced collaboration. Employees can work on shared documents and projects simultaneously, even from different locations, increasing teamwork and productivity.

Support better security and backup. Organizations can implement centralized security controls and regular data backups, protecting information from loss or unauthorized access.

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036/2

INFORMATION AND COMPUTER STUDIES 2

(ACTUAL PRACTICAL)

(For Both School and Private Candidates)

Duration: 3 Hours

SOLUTIONS

Year: 2025

Instructions

1. This paper consists of three (3) questions.
2. Answer any two (2) questions. Each question carries 25 marks
3. Save your work on the desktop using the folder named by your examination number
4. Communication devices and any unauthorised materials are **not** allowed in the examination room.
5. Write your **Examination Number** on every page of your answer booklet(s)

1. Mr. Majao manages his Shoes Shop records using Microsoft Access database. The database stores customer records including contacts and purchases. He uses the database as the marketing strategy for facilitating him in awarding his royal customers. The customer is awarded a free pair of shoes if he buys 10 pairs. Study carefully the following tables which are included in the database and answer the questions that follow:

Customers

CustID	FirstName	Surname	Phone#
1	Jona	Muliye	0713345610
2	Mary	Kolida	0674234012
3	Kilo	Kilosa	0782919293
4	Juju	Moses	0761839099
5	Marcelina	Mosi	0651828280
6	Moneylisa	Pesa	0621345334
7	Joy	Majiji	0765080911

Purchases

PurchaseID	CustID	DatePurchase	Shoe_Category	Num_Shoes	Amount
1	1	3/8/2023	ladies	1	35,000.00
2	1	4/6/2023	ladies	3	30,000.00
3	1	5/5/2023	kids	4	60,000.00
4	1	4/7/2023	Men	2	48,000.00
5	2	8/2/2023	ladies	2	30,000.00
6	2	5/23/2023	kids	1	15,000.00
7	3	6/23/2023	men	1	39,000.00
8	3	6/18/2023	kids	2	28,000.00
9	4	3/26/2023	men	2	56,000.00
10	5	4/6/2023	kids	3	33,000.00
11	5	5/20/2023	ladies	2	40,000.00
12	6	4/23/2016	men	1	35,000.00
13	7	5/23/2023	ladies	1	18,000.00
14	7	5/27/2023	ladies, men	4	98,000.00
15	7	6/20/2023	kids, ladies, men	3	85,000.00
16	7	6/28/2023	kids	2	30,000.00

- (a) Create the database for Mr. Majao Shoes Shop and save it as "MajaoShoesDB".
- (b) Create the given Customers and Purchases tables within MajaoShoesDB.
- (c) Use Form Wizard to create a form that enables Mr. Majao to enter customers and purchases data using one form. Use the form created to enter the sample data provided. Save the form as "Customers Purchasesform".
- (d) Use Customers and Purchases tables to create a one-to-many relationship between the two tables.
- (e) Add mathematical formula to the form created in 1 (c) that enables Majao to view the total amount of money that each customer spent and the number of pairs bought.
- (f) Create a query that enables to check customers' purchase details. The details include customer ID, first name, surname, Phone number, date purchased, pairs of shoes and amount paid for each day. Save the query as "customerspurchases".
- (g) Create a query which enables to view customer's first name, surname, dates and shoes category for customers who bought ladies or kids shoes. Save the query as "bought ladies or kids shoes".
- (h) Create a parameter query that enables to search for any customer using customer's surname and display his purchase records. Save the query as "search a customer by surname".
- (i) Create a report that search and prints customers' purchase records by their surnames. Save the report as "Report of Customers by surname".
- (j) Create a report that will enable Majao to print records of all customers who bought men shoes. Save the report as "Men shoes only report".

Database Structure and Relationships

Tables

1. Customers Table

- Primary Key: CustID
- Fields: FirstName, Surname, Phone#

2. Purchases Table

- Primary Key: PurchaseID
- Foreign Key: CustID (links to Customers)
- Fields: DatePurchase, Shoe_Category, Num_Shoes, Amount

Relationship:

- **One-to-Many:** One customer can have multiple purchases.
- **CustID** in Customers → CustID in Purchases

Form:

- **Customers Purchasesform** combines Customers and Purchases tables.
- Allows input for both customer details and purchases on a single form.
- Includes **mathematical controls** to calculate total amount spent and total shoes purchased.

Queries:

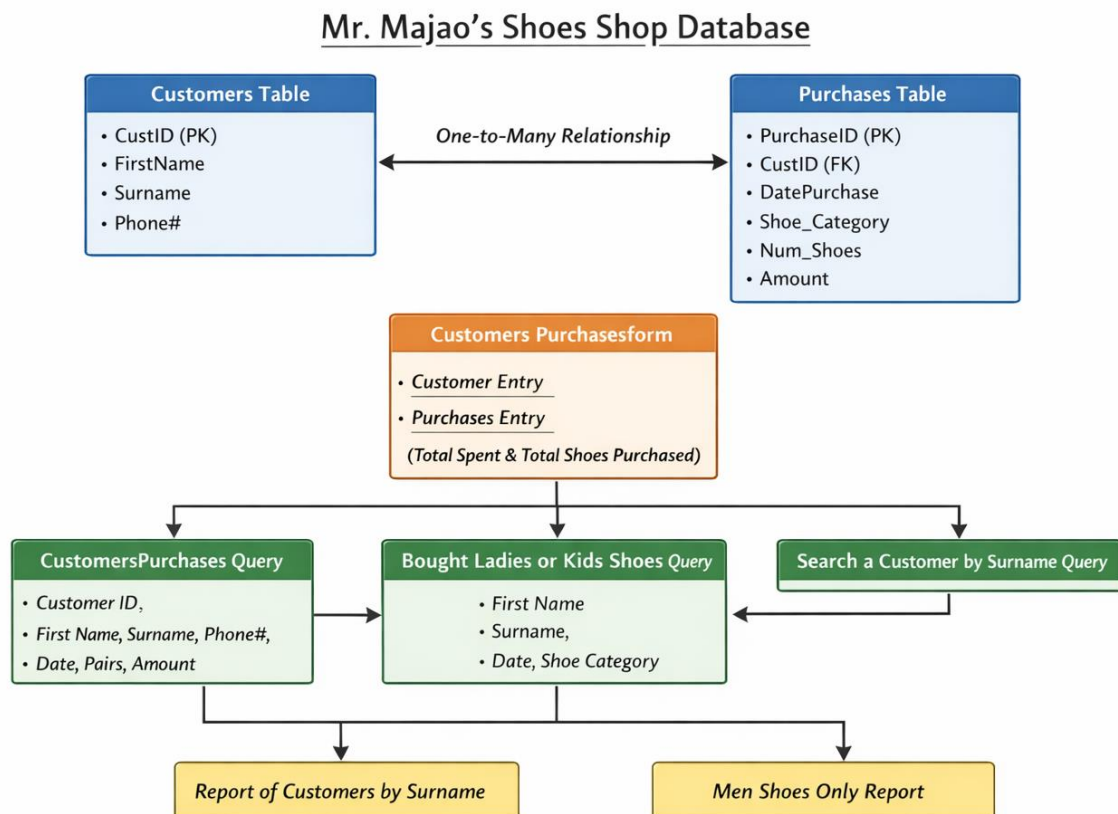
- **customerspurchases** → displays all purchase details with customer information.

- **bought ladies or kids shoes** → filters purchases for ladies or kids shoes.
- **search a customer by surname** → parameter query to search a specific customer.

Reports:

- **Report of Customers by surname** → prints purchases grouped by surname.
- **Men shoes only report** → prints purchases where Shoe_Category contains “Men”.

Diagram Representation



Explanation of Diagram:

- **Customers → Purchases:** Shows the one-to-many relationship.
- **Queries:** Pull data from both tables to filter or summarize information.
- **Reports:** Use the queries as sources to generate formatted, printable output.

2. Umoja Technical College offers different technical courses in its campuses allocated at different areas in Tanzania. To make teachers and students aware of all monthly-scheduled events, the principal of the college asked you to prepare online calendar to the college website.

(a) Create the following Interface by using basic HTML codes.

Hint: Use table with border 0, height 300 and width 300 to create the Interface and to arrange the monthly buttons.

Page descriptions

- The heading "UMOJA CALENDAR" should be colored in red with the font type "Gabriola" and text size 20.
- The size of the horizontal line should be 300.

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Umoja Calendar</title>
```

```
</head>
```

```
<body>
```

```
<h1 style="color: red; font-family: Gabriola; font-size: 20px;">UMOJA  
CALENDAR</h1>
```

```
<hr width="300" align="left">
```

```
<table border="0" height="300" width="300">
```

<tr>

<td><button>January</button></td>

<td><button>February</button></td>

<td><button>March</button></td>

</tr>

<tr>

<td><button>April</button></td>

<td><button>May</button></td>

<td><button>June</button></td>

</tr>

<tr>

<td><button>July</button></td>

<td><button>August</button></td>

<td><button>September</button></td>

</tr>

<tr>

<td><button>October</button></td>

<td><button>November</button></td>

<td><button>December</button></td>

</tr>

</table>

</body>

</html>

(b) Activate the button "May" in the interface created in 2 (a) so that when the user clicks, it displays the following page.

HINT:

- Use Microsoft word to create the circle and rectangle shapes to the dates "1" and "4" respectively.
- Use the borderless table to arrange the dates.

Page descriptions

- The heading "May 2025" should be in level 2.
- The size of the horizontal line after the heading should be 300 and aligned at centre.
- The background and line colour of the circle shape should be orange, accent = 2, lighter = 80% and the line colour should be blue.
- The line colour of the rectangle shape in the 4th date should be blue.
- The height and width of the table required to arrange the dates should be 350 for each.

(c) Develop the HTML codes that activate the 1st date in the circle shape created in 2(b) such that the following page will be displayed when the user clicks it.

1st May 2025 Worker's day

Page descriptions

- The font type, colour and size of the text "Worker's day" should be "Lucida handwriting", "white" and 13 respectively and the background color should be green.

(d) Construct HTML codes that display the "current date" when the user hangs the mouse on the rectangle button labeled with number "4" in the interface created in 2 (b).

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>May 2025</title>
```

```
<style>
```

```
.circle {
```

```
    background-color: #fbe5d6; /* Orange Accent 2 Lighter 80% */
```

```
    border: 2px solid blue;
```

```
    border-radius: 50%;
```

```
    padding: 5px;
```

```
    display: inline-block;
```

```
}
```

```
.rect {
```

```
    border: 2px solid blue;
```

```

padding: 5px;

display: inline-block;

}

</style>

</head>

<body>

<h2 align="center">May 2025</h2>

<hr width="300" align="center">

<table border="0" height="350" width="350" align="center">

<tr><th>Sun</th><th>Mon</th><th>Tue</th><th>Wed</th><th>Thur</th><th>Fri</t
h><th>Sat</th></tr>

<tr>

<td></td><td></td><td></td><td></td>

<td><a href="workersday.html" class="circle">1</a></td>

<td>2</td><td>3</td>

</tr>

<tr>

<td><span class="rect" title="04/02/2026">4</span></td>

<td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td>

```

</tr>

</table>

</body>

</html>